



Dr. Kapil Saraswat

Researcher & Teacher
Sharma Classes, Amroha, UP, India
Institute of Physics,
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Vidwan Profile

in LinkedIn Profile

EDUCATION

- **Banaras Hindu University, Varanasi, India** 2018
Ph.D. Physics
- **University of Delhi, Delhi, India** 2009
M.Sc. Physics
- **M. J. P. Rohilkhand University, Bareilly, India** 2007
B.Sc. Physics & Mathematics
- **Kundan Model Intermediate College, Amroha, Uttar Pradesh, india** 2004
12th, Board of HighSchool And Intermediate Education Uttar Pradesh, Prayagraj, india
- **Kundan Model Intermediate College, Amroha, Uttar Pradesh, india** 2002
10th, Board of HighSchool And Intermediate Education Uttar Pradesh, Prayagraj, india

EXPERIENCE

- **Sharma Classes, Amroha, Uttar Pradesh, India** July 2022 - Present
Teacher
- **Institute of Physics, Academia Sinica, Taipei, Taiwan** March 2020 - May 2022
Post Doctoral Fellow
- **Kumaun University, Nainital, India** January 2019 - February 2020
Post Doctoral Fellow

AREAS OF SCIENTIFIC INTEREST

- Relativistic heavy-ion collisions at ultra-relativistic energies
- Neutrino interaction with matter at low, intermediate and high energies
- Data analysis of neutrino-nucleus scattering
- Geant4 based simulation studies for various physics processes

TECHNICAL SKILLS

Languages : C++ & Elementary Python
Frameworks : CERN Root & Pyroot
Scientific Tools : Mathematica, Latex & GNU Scientific Libraray
Operating System : Mac & Linux

RESEAECH PUBLICATIONS

Publications in international journals : 20
Publications in national journals : 1
Submitted in international Journals : 3
Publications in conference proceedings : 37
Chapters in books : 3

ACHIEVEMENTS

- **MOST Postdoctoral Fellowship** Awarded By Ministry of Science And Technology, Taiwan December 2021
- **D. S. Kothari Postdoctoral Fellowship** Awarded by University Grant Commission, New Delhi, India December 2018
- **UGC NET-JRF** Awarded by University Grant Commission, New Delhi, India June 2010
- **GATE** Awarded by IIT, Guwahati, India March 2010

Dr. Kapil Saraswat

Curriculum Vitae

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Whitefield, Bengaluru, Karnataka, India
Subodh Nagar, Amroha, Uttar Pradesh, India.
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Work Experience

- July 2022 – **Teacher at Sharma Classes, Amroha, UP & Researcher.**
Present
- March 2020 – **Postdoctoral Fellow**, *Taiwan Experiment On Neutrino (TEXONO) & China Dark Matter Experiment (CDEX)*, Institute of Physics, Academia Sinica, Taipei, Taiwan.
May 2022
- January 2019 – **Postdoctoral Fellow**, Department of Physics, Kumaun University, Nainital, India.
– February 2020

Education

- 2012–2018 **Ph.D.**, *Physics*, Banaras Hindu University, Varanasi, India.
Thesis Title : Study of Strongly Interacting Matter using Light and Heavy Hadron Production in Heavy Ion Collisions.
○ **Thesis Supervisor** : Prof. Venktesh Singh^{1,2} and Prof. Prashant Shukla^{3,4}.
○ ¹Department of Physics, Institute of Science, Banaras Hindu University, Varanasi, India.
○ ²Department of Physics, School of Physical Sciences, Central University of South Bihar, Gaya, India.
○ ³Nuclear Physics Division, Bhabha Atomic Research Centre, Mumbai, India.
○ ⁴Homi Bhabha National Institute, Mumbai, India.
- 2007–2009 **M.Sc.**, *Physics*, University of Delhi, Delhi, India.
- 2004–2007 **B.Sc.**, *Physics & Mathematics*, M. J. P . Rohilkhand University, Bareilly, India.
- 2002–2004 **Intermediate (12th)**, *Science Stream*, Board of High School And Intermediate Education Uttar Pradesh, Prayagraj, India.
- 2000–2002 **High School (10th)**, *Science Stream*, Board of High School And Intermediate Education Uttar Pradesh, Prayagraj, India.

Areas of Scientific Interest

- Heavy quark production in proton-proton, proton-nucleus and nucleus-nucleus collisions at ultra-relativistic energies.
- Heavy quark energy loss in Quark-Gluon Plasma(QGP) medium.
- Particle production mechanism in proton-proton, proton-nucleus and nucleus-nucleus collisions at ultra-relativistic energies.

- Neutrino interaction with nucleon/nucleus at low, intermediate and high energies.
- Data analysis of neutrino-nucleus elastic scattering.
- Ge detector based background simulation studies for Neutrinoless double beta decay and Coherent neutrino-nucleus elastic scattering.

Awards And Merits

- 2021 **MOST Postdoctoral Fellowship.**
Awarded By : Ministry of Science and Technology, Taiwan
- 2019 **D. S. Kothari Postdoctoral Fellowship.**
Awarded By : University Grants Commission, New Delhi, India
- 2014 **Senior Research Fellowship.**
Awarded By : University Grants Commission, New Delhi, India
- 2012 **Junior Research Fellowship.**
Awarded By : University Grants Commission, New Delhi, India

National Level Exams

- December 2009** **JRF-NET**, Joint CSIR-UGC National Eligibility Test.
- March 2010** **GATE**, Graduate Aptitude Test In Engineering.
- March 2010** **JEST**, Joint Entrance Screening Test.

Technical Skills

- Programming Language** C++
- Analysis Software** CERN ROOT
- Scientific Tools** Mathematica, Latex, and GNU Scientific Libraray
- Simulation** GEANT4 Based Simulation Package
- Operating System** Mac and Linux (Fedora and Ubuntu)

List of Publications

1. **Title :** Constraining heavy quark energy loss using B and D meson measurements in heavy ion collision at RHIC and LHC energies
Authors : **Kapil Saraswat**, Prashant Shukla, and Venktesh Singh.
Journal : *Nucl. Phys. A* **943** (2015) 83, [[arXiv:1507.06742 \[nucl-th\]](#)].
2. **Title :** Coherent pion production in neutrino-nucleus scattering.
Authors : **Kapil Saraswat**, Prashant Shukla, Vineet Kumar, and Venktesh Singh.
Journal : *Phys. Rev. C* **93** (2016) 035504, [[arXiv:1602.07820 \[hep-ph\]](#)].
3. **Title :** Energy loss of heavy quarks and B and D meson spectra in PbPb collisions at LHC energies.
Authors : **Kapil Saraswat**, Prashant Shukla, Vineet Kumar, and Venktesh Singh.
Journal : *Nucl. Phys. A* **961** (2017) 169, [[arXiv:1702.05733 \[hep-ph\]](#)].
4. **Title :** Strange hadron production in pp, pPb and PbPb collisions at LHC energies.
Authors : **Kapil Saraswat**, Prashant Shukla, Vineet Kumar, and Venktesh Singh.
Journal : *Eur. Phys. J. A* **53** (2017) 84, [[arXiv:1702.05734 \[nucl-th\]](#)].
5. **Title :** Charged current quasi elastic scattering of muon neutrino with nuclei.
Authors : **Kapil Saraswat**, Prashant Shukla, Vineet Kumar, and Venktesh Singh.
Journal : *Indian J. Phys.* **92** (2018) 249, [[arXiv:1606.06916 \[hep-ph\]](#)].
6. **Title :** Transverse momentum spectra of hadrons in high energy pp and heavy ion collisions.
Authors : **Kapil Saraswat**, Prashant Shukla, and Venktesh Singh.
Journal : *J. Phys. Comm.* **2**(2018) 035003, [[arXiv:1706.04860 \[hep-ph\]](#)].
7. **Title :** Charged current quasi elastic scattering of muon anti-neutrino off ^{12}C .
Authors : Deepika Grover, **Kapil Saraswat**, Prashant Shukla, and Venktesh Singh.
Journal : *Chin. Phys. C* **42** (2018) 123104, [[arXiv:1807.08911 \[hep-ph\]](#)].
8. **Title :** Charged current deep inelastic scattering of ν_μ off ^{56}Fe .
Authors : Deepika Grover, **Kapil Saraswat**, Prashant Shukla, and Venktesh Singh.
Journal : *Phys. Rev. C* **98** (2018) 065503, [[arXiv:1808.00287 \[hep-ph\]](#)].
9. **Title :** Understanding partonic energy loss from measured light charged particles and jets in PbPb collisions at LHC energies.
Authors : Prashant Shukla, and **Kapil Saraswat**.
Journal : *J.Phys. G* **47** (2020) 125103, [[arXiv:2105.06364 \[hep-ph\]](#)].
10. **Title :** Medium effects of charged particles in Xe+Xe collisions at $\sqrt{s_{\text{NN}}} = 5.44$ TeV using modified Tsallis distribution.
Authors : Pramod Kumar, P. K. Khandai, **Kapil Saraswat**, and V. Singh.

- Journal :** [Int.J.Mod.Phys. A 36 \(2021\) 2150059.](#)
11. **Title :** Studies of quantum-mechanical coherency effects in neutrino-nucleus elastic scattering.
Authors : V. Sharma, L. Singh, H. T. Wong, M. Agartioglu, J. W. Chen, M. Deniz, S. Kerman, H. B. Li, C. P. Liu, **Kapil Saraswat**, M. K. Singh, and V. Singh (Texono Collaboration).
Journal : [Phys. Rev. D103 \(2021\) 092002](#), [\[arXiv:2010.06810 \[hep-ex\]\]](#).
 12. **Title :** Theoretical analysis of p_T spectra of light-flavor hadrons in $p+p$ collisions at $\sqrt{s} = 7$ TeV under differential and single freeze-out scenarios.
Authors : Pramod Kumar, P. K. Khandai, **Kapil Saraswat**, and V. Singh.
Journal : [Int.J.Mod.Phys.A 36 \(2021\) 2150160.](#)
 13. **Title :** Studies of earth shielding effect to direct dark matter searches at the China Jinping Underground Laboratory.
Authors : Z. Z. Liu, **Kapil Saraswat** *et al.* (CDEX Collaboration).
Journal : [Phys. Rev. D 105 \(2022\) 052005](#), [\[arXiv:2111.11243 \[hep-ex\]\]](#).
 14. **Title :** Study of Charm and Bottom Quark Energy Loss and Associated Meson R_{AA} Spectra in Proton-Lead Collisions at $\sqrt{s_{NN}} = 5.02$ TeV.
Authors : **Kapil Saraswat**, Deependra Singh Rawat and H.C. Chandola.
Journal : [Nucl. Phys. A 1022 \(2022\) 122441.](#)
 15. **Title :** Search for Neutrinoless Double-Beta Decay of ^{76}Ge with a Natural Broad Energy Germanium Detector.
Authors : W.H. Dai, **Kapil Saraswat** *et al.* (CDEX Collaboration).
Journal : [Phys. Rev. D 106 \(2022\) 032012](#), [\[arXiv:2205.10718 \[nucl-ex\]\]](#).
 16. **Title :** Constraints on sub-GeV Dark Matter Boosted by Cosmic Rays from CDEX-10 Experiment at the China Jinping Underground Laboratory.
Authors : R. Xu, **Kapil Saraswat** *et al.* (CDEX Collaboration).
Journal : [Phys. Rev. D 106 \(2022\) 052008](#), [\[arXiv:2201.01704 \[hep-ex\]\]](#).
 17. **Title :** Constraints on Sub-GeV Dark Matter-Electron Scattering from the CDEX-10 Experiment.
Authors : Z. Y. Zhang, L. T. Yang, **Kapil Saraswat** *et al.* (CDEX Collaboration).
Journal : [Phys. Rev. Lett. 129 \(2022\) 221301](#), [\[arXiv:2206.04128 \[hep-ex\]\]](#).
 18. **Title :** Potential of thermal protection improvements in large-scale cryostat development.
Authors : M. K. Singh, V. Singh, **K. Saraswat**, and D. Singh.
Journal : [International Journal of Heat and Mass Transfer 236 \(2025\) 126254.](#)

19. **Title :** New Limits on Coherent Neutrino Nucleus Elastic Scattering Cross Section at the Kuo-Sheng Reactor Neutrino Laboratory.
Authors : S. Karmakar, M.K. Singh, H.T. Wong, **Kapil Saraswat**, *et al.* (TEXONO, Collaboration).
Journal : [Phys. Rev. Lett. **134** \(2025\) 121802.](#), [\[arXiv:2411.18812 \[nucl-ex\]\]](#).
20. **Title :** Dark Matter Annual Modulation Analysis with Combined Nuclear and Electron Recoil Channels.
Authors : H. B. Li, M.K. Pandey, H.T. Wong, **Kapil Saraswat**, *et al.* (TEXONO, Collaboration).
Journal : [Phys. Rev. D **111** \(2025\) 083035.](#), [\[arXiv:2412.04916 \[nucl-ex\]\]](#).
21. **Title :** Impact of theoretical constraints in the sensitivity estimation for neutrinoless double beta decay.
Authors : M. K. Singh, S. Karmakar, H.T. Wong, **Kapil Saraswat**, *et al.*
Journal : [Int. J. Mod. Phys. A **40** \(2025\) 2550050.](#)
22. **Title :** Medium effects of charged-hadron production in pPb, and PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV using modified Tsallis distribution.
Authors : **Kapil Saraswat**, Prashanta Kumar Khandai, and Venktesh Singh.
Journal : Submitted In [Progress of Theoretical and Experimental Physics \(2024\)](#), [\[arXiv : 2405.12231 \[hep-ph\]\]](#).
23. **Title :** Phenomenological study of the charged particles production in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV.
Authors : **Kapil Saraswat**, Deependra Singh Rawat, Akash Pandey, Venktesh Singh, and H. C. Chandola.
Journal : Submitted In [International Journal of Modern Physics A \(2024\)](#), [\[arXiv : 2407.14182 \[hep-ph\]\]](#).
24. **Title :** Study of medium effects on the ATLAS measured data of charged-hadron production in XeXe collisions at $\sqrt{s_{NN}} = 5.44$ TeV using modified Tsallis distribution.
Authors : **Kapil Saraswat**, Prashanta Kumar Khandai, and Venktesh Singh.
Journal : To Be Communicated In [Advances in High Energy Physics \(2024\)](#) .

List of Papers Published in Conference and Proceedings

1. **Conference :** 57th DAE BRNS Symposium on Nuclear Physics 2012 at Department of Physics and Astrophysics, University of Delhi, Delhi, India.
Title : China Jin-Ping Deep Underground Laboratory: Status and Plan.
Authors : Arun Kumar Soma, Manoj Kumar Singh, Lakhwinder Singh, Vivek Sharma, **Kapil Saraswat**, Venktesh Singh, V. S. Subrahmanyam, Henry Tsz-king Wong.
Journal : [DAE Symp. Nucl. Phys. 57 \(2012\) 640.](#)
2. **Conference :** 57th DAE BRNS Symposium on Nuclear Physics 2012 at Department of Physics and Astrophysics, University of Delhi, Delhi, India.
Title : Extracting Physics from P+ PCGe Detector near Noise Edge.
Authors : Venktesh Singh, Manoj Kumar Singh, Lakhwinder Singh, Vivek Sharma, Nishant Singh Chouhan, **Kapil Saraswat**, V. S. Subrahmanyam, Henry Tsz-king Wong.
Journal : [DAE Symp. Nucl. Phys. 57 \(2012\) 866.](#)
3. **Conference :** 57th DAE BRNS Symposium on Nuclear Physics 2012 at Department of Physics and Astrophysics, University of Delhi, Delhi, India.
Title : Study of Surface Resistivity of Resistive Plate Chamber Detectors.
Authors : Manoj Kumar Jaiswal, Vivek Sharma, Nishant Singh Chouhan, **Kapil Saraswat**, Venktesh Singh, Vajapeyula Srinivasa Subrahmanyam.
Journal : [DAE Symp. Nucl. Phys. 57 \(2012\) 968.](#)
4. **Conference :** 57th DAE BRNS Symposium on Nuclear Physics 2012 at Department of Physics and Astrophysics, University of Delhi, Delhi, India.
Title : Trigger Detector Study for Resistive Plate Chamber Detectors.
Authors : **Kapil Saraswat**, Venktesh Singh, Vajapeyula Srinivasa Subrahmanyam.
Journal : [DAE Symp. Nucl. Phys. 57 \(2012\) 970.](#)
5. **Conference :** 58th DAE BRNS International Symposium on Nuclear Physics 2013 at Bhabha Atomic Research Centre, Mumbai, India.
Title : Study of cosmic ray muons momentum and charge asymmetry spectra.
Authors : V. Sharma, **K. Saraswat**, V. Singh, and V. S. Subrahmanyam.
Journal : [DAE Symp. Nucl. Phys. 58 \(2013\) 814.](#)
6. **Conference :** 58th DAE BRNS International Symposium on Nuclear Physics 2013 at Bhabha Atomic Research Centre, Mumbai, India.
Title : Study of VECC Prototype INO ICAL using cosmic ray muon and g4beamline.
Authors : **Kapil Saraswat**, Lakhwinder Singh, Prashata Kumar Khandai, Manoj Kumar Jaiswal, Lovika Moudgil, O. R. Rahul, Vikas Dixit, Venktesh Singh, V. S. Subrahmanyam.
Journal : [DAE Symp. Nucl. Phys. 58 \(2013\) 972.](#)

7. **Conference :** ^{59th}DAE BRNS Symposium on Nuclear Physics 2014 at Department Of Physics, Banaras Hindu University, Varanasi, India.
Title : Constraining Heavy Quark Energy Loss Using B and D Mesons in Heavy Ion Collision.
Authors : **Kapil Saraswat**, Prashant Shukla, and Venktesh Singh.
Journal : **Journal Ref. :** [DAE Symp. Nucl. Phys. **59** \(2014\) 766.](#)

8. **Conference :** ^{59th}DAE BRNS Symposium on Nuclear Physics 2014 at Department Of Physics, Banaras Hindu University, Varanasi, India.
Title : Interaction of neutrinos with matter in low and intermediate energy region.
Authors : **Kapil Saraswat**, Prashant Shukla, Vineet Kumar, and Venktesh Singh.
Journal : [DAE Symp. Nucl. Phys. **59** \(2014\) 798.](#)

9. **Conference :** ^{60th}DAE BRNS Symposium on Nuclear Physics 2015 at Sri Sathya Sai Institute of Higher Learning, Prasanthi Nilayam, India.
Title : Heavy quark energy loss in PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV.
Authors : **Kapil Saraswat**, Prashant Shukla, and Venktesh Singh.
Journal : [DAE Symp. Nucl. Phys. **60** \(2015\) 730.](#)

10. **Conference :** ^{60th}DAE BRNS Symposium on Nuclear Physics 2015 at Sri Sathya Sai Institute of Higher Learning, Prasanthi Nilayam, India.
Title : Coherent pion production in ν -Nucleus interaction.
Authors : **Kapil Saraswat**, Prashant Shukla, Vineet Kumar, and Venktesh Singh.
Journal : [DAE Symp. Nucl. Phys. **60** \(2015\) 848.](#)

11. **Conference :** ^{7th}International Conference on Physics and Astrophysics of Quark Gluon Plasma (ICPAQGP) 2015 at Variable Energy Cyclotron Center, Kolkata, India.
Title : Energy loss of B and D mesons in PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV.
Authors : **Kapil Saraswat**, Prashant Shukla, and Venktesh Singh.
Journal : [PoS ICPAQGP2015 \(2015\) 082, \[arXiv:1511.05349 \\[hep-ph\\].\]\(#\)](#)

12. **Conference :** ^{61th}DAE BRNS Symposium on Nuclear Physics 2016 at Saha Institute of Nuclear Physics, Kolkata, India.
Title : Strange hadron production in high multiplicity pp collision at $\sqrt{s} = 7$ TeV.
Authors : **Kapil Saraswat**, Prashant Shukla, Vineet Kumar, and Venktesh Singh.
Journal : [DAE Symp. Nucl. Phys. **61** \(2016\) 780.](#)

13. **Conference :** ^{61th}DAE BRNS Symposium on Nuclear Physics 2016 at Saha Institute of Nuclear Physics, Kolkata, India.
Title : Quasi elastic scattering of neutrino with nuclei.
Authors : **Kapil Saraswat**, Prashant Shukla, Vineet Kumar, and Venktesh Singh.
Journal : [DAE Symp. Nucl. Phys. **61** \(2016\) 866.](#)

14. **Conference :** XXII DAE BRNS HIGH ENERGY PHYSICS SYMPOSIUM 2016 at University of Delhi, Delhi, India.
Title : Charged and Neutral coherent pion production in neutrino-nucleus scattering.
Authors : **Kapil Saraswat**, Prashant Shukla, Vineet Kumar, and Venktesh Singh.
Journal : [Springer Proc. Phys. **203** \(2018\) 697](#), [[arXiv:1805.11836 \[hep-ph\]](#)].
15. **Conference :** XXII DAE BRNS HIGH ENERGY PHYSICS SYMPOSIUM 2016 at University of Delhi, Delhi, India.
Title : Freeze out of strange hadrons in pp, pPb and PbPb collisions at LHC energies.
Authors : **Kapil Saraswat**, Prashant Shukla, Vineet Kumar and Venktesh Singh.
Journal : [Springer Proc. Phys. **203** \(2018\) 701](#), [[arXiv:1805.11832 \[hep-ph\]](#)].
16. **Conference :** ^{62th} DAE BRNS Symposium on Nuclear Physics 2017 at Thapar University, Patiala, India
Title : Heavy quark energy loss and heavy meson spectra in heavy ion collisions at LHC energies.
Authors : **Kapil Saraswat**, Prashant Shukla, Vineet Kumar, and Venktesh Singh.
Journal : [DAE Symp. Nucl. Phys. **62** \(2017\) 850](#).
17. **Conference :** ^{62th} DAE BRNS Symposium on Nuclear Physics 2017 at Thapar University, Patiala, India
Title : Charged particle production in pp and heavy ion collisions.
Authors : **Kapil Saraswat**, Prashant Shukla, Vineet Kumar, and Venktesh Singh.
Journal : [DAE Symp. Nucl. Phys. **62** \(2017\) 856](#).
18. **Conference :** ^{62th} DAE BRNS Symposium on Nuclear Physics 2017 at Thapar University, Patiala, India
Title : Charged current deep inelastic scattering of muon neutrino with Iron nuclei.
Authors : Deepika Grover, **Kapil Saraswat**, Prashant Shukla, and Venktesh Singh.
Journal : [DAE Symp. Nucl. Phys. **62** \(2017\) 972](#).
19. **Conference :** ^{62th} DAE BRNS Symposium on Nuclear Physics 2017 at Thapar University, Patiala, India
Title : Study of strongly interacting matter using light and heavy hadron production in heavy ion collisions.
Authors : **Kapil Saraswat**.
Journal : [DAE Symp. Nucl. Phys. **62** \(2017\) 1202](#).
20. **Conference :** ^{63th} DAE BRNS International Symposium on Nuclear Physics 2018 at Bhabha Atomic Research Centre, Mumbai, India.
Title : Identified charged particle production in proton-proton collisions at $\sqrt{s} = 13$ TeV.
Authors : Pramod Kumar, **Kapil Saraswat**, Prashant Shukla and Venktesh Singh.

- Journal :** [DAE Symp. Nucl. Phys. 63 \(2018\) 966.](#)
21. **Conference :** ^{63th}DAE BRNS International Symposium on Nuclear Physics 2018 at Bhabha Atomic Research Centre, Mumbai, India.
Title : Nuclear modification factor and energy loss of charged particles and jets in PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV.
Authors : **Kapil Saraswat**, Prashant Shukla, and Venktesh Singh.
Journal : [DAE Symp. Nucl. Phys. 63 \(2018\) 980.](#)
22. **Conference :** ^{63th}DAE BRNS International Symposium on Nuclear Physics 2018 at Bhabha Atomic Research Centre, Mumbai, India.
Title : Quasi elastic scattering of anti-muon neutrinos off ^{12}C .
Authors : Deepika Grover, **Kapil Saraswat**, Prashant Shukla, and Venktesh Singh.
Journal : [DAE Symp. Nucl. Phys. 63 \(2018\) 1032.](#)
23. **Conference :** ^{64th}DAE BRNS Symposium on Nuclear Physics 2019 at Department Of Physics, University Of Lucknow, Lucknow, India.
Title : Heavy quark energy loss and heavy meson spectra in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV.
Authors : **Kapil Saraswat**, Prashant Shukla, and H. C. Chandola.
Journal : [DAE Symp. Nucl. Phys. 64 \(2019\) 716.](#)
24. **Conference :** ^{64th}DAE BRNS Symposium on Nuclear Physics 2019 at Department Of Physics, University Of Lucknow, Lucknow, India.
Title : Centrality dependence of Identified charged Particle Production in Pb+Pb collisions at $\sqrt{s} = 2.76$ TeV.
Authors : Pramod Kumar, **Kapil Saraswat**, and Venktesh Singh.
Journal : [DAE Symp. Nucl. Phys. 64 \(2019\) 750.](#)
25. **Conference :** ^{64th}DAE BRNS Symposium on Nuclear Physics 2019 at Department Of Physics, University Of Lucknow, Lucknow, India.
Title : Deuteron production in Au + Au collisions at RHIC energies.
Authors : P. K Khandai, and **Kapil Saraswat**.
Journal : [DAE Symp. Nucl. Phys. 64 \(2019\) 778.](#)
26. **Conference :** ^{65th} DAE BRNS Symposium on Nuclear Physics 2021 at Bhabha Atomic Research Centre, Mumbai, India.
Title : Study of medium modifications in Xe+Xe collisions at $\sqrt{s_{NN}} = 5.44$ TeV.
Authors : P.K. Khandai, P. Kumar, **K. Saraswat**, and V. Singh.
Journal : [DAE Symp. Nucl. Phys. 65 \(2021\) 632.](#)

27. **Conference :** ^{65th} DAE BRNS Symposium on Nuclear Physics 2021 at Bhabha Atomic Research Centre, Mumbai, India.
Title : Effect of energy loss on nuclear modification factor of heavy mesons in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV.
Authors : **Kapil Saraswat**, Deependra Singh Rawat, and H. C. Chandola.
Journal : [DAE Symp. Nucl. Phys. 65 \(2021\) 676.](#)
28. **Conference :** ^{65th} DAE Symposium on Nuclear Physics 2021 at Bhabha Atomic Research Centre, Mumbai, India.
Title : Characteristic Behavior of Color Flux-Tubes in Dual QCD.
Authors : Deependra Singh Rawat, H. C. Chandola, and **Kapil Saraswat**.
Journal : [DAE Symp. Nucl. Phys. 65 \(2021\) 710.](#)
29. **Conference :** ^{66th} DAE Symposium on Nuclear Physics 2022 at Cotton University, Guwahati, Assam, India.
Title : Implications of DMHD in Color Flux Tube Dynamics of QCD.
Authors : Deependra Singh Rawat, H. C. Chandola, **Kapil Saraswat**, and Mehul Manu.
Journal : [DAE Symp. Nucl. Phys. 66 \(2022\) 1050.](#)
30. **Conference :** ^{66th} DAE Symposium on Nuclear Physics 2022 at Cotton University, Guwahati, Assam, India.
Title : Theoretical uncertainty in the sensitivity projection for neutrinoless double beta decay.
Authors : Manoj Kumar Singh, Henry Tsz-King Wong, **Kapil Saraswat**, Shashank Mishra, Damini Singh, Saurabh Shukla, and Venktesh Singh.
Journal : [DAE Symp. Nucl. Phys. 66 \(2022\) 1069.](#)
31. **Conference :** ^{66th} DAE Symposium on Nuclear Physics 2022 at Cotton University, Guwahati, Assam, India.
Title : Identification of physics signal-events near threshold in p-type point-contact Germanium detectors.
Authors : Manoj Kumar Singh, Henry Tsz-King Wong, **Kapil Saraswat**, Vivek Sharma, and Venktesh Singh.
Journal : [DAE Symp. Nucl. Phys. 66 \(2022\) 1178.](#)
32. **Conference :** ^{67th} DAE Symposium on Nuclear Physics 2023 at Indian Institute of Technology, Indore, India.
Title : Charged particles production in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV using modified Tsallis distribution.
Authors : Akash Pandey, **Kapil Saraswat**, Deependra Singh Rawat, Manoj Kumar Singh, Vivek Sharma, Damini Singh, Pramod Kumar, Prashanta Kumar Khandai, Venktesh Singh.

- Journal :** [DAE Symp. Nucl. Phys. 67 \(2023\) 1079.](#)
33. **Conference :** *67th* DAE Symposium on Nuclear Physics 2023 at Indian Institute of Technology, Indore, India.
- Title :** The contribution of arrangement of radiation shields in the thermal efficiency of multilayer insulation technique.
- Authors :** Dr. Damini Singh, Manoj Kumar Singh, **Kapil Saraswat**, Ankur Chaubey, Avijit Kumar Ganguly, Venktesh Singh, and Sonika.
- Journal :** [DAE Symp. Nucl. Phys. 67 \(2023\) 1315.](#)
34. **Conference :** *67th* DAE Symposium on Nuclear Physics 2023 at Indian Institute of Technology, Indore, India.
- Title :** Impact of reactor-associated high energy background in the sub-keV energy region of the TEXONO experiment.
- Authors :** Manoj Kumar Singh, Shuvadeep Karmakar, H.T. Wong, **Kapil Saraswat**, Venktesh Singh.
- Journal :** [DAE Symp. Nucl. Phys. 67 \(2023\) 1317.](#)
35. **Conference :** *68th* DAE Symposium on Nuclear Physics 2024 at Indian Institute of Technology, Roorkee, India.
- Title :** Medium effects of charged-hadron production in Xe-Xe collisions at $\sqrt{s_{NN}} = 5.02$ TeV using modified Tsallis distribution.
- Authors :** **Kapil Saraswat**, Prashanta Kumar Khandai, Deependra Singh Rawat, Damini Singh, Manoj Kumar Singh, Venktesh Singh.
- Journal :** [DAE Symp. Nucl. Phys. 68 \(2024\) 995.](#)
36. **Conference :** *68th* DAE Symposium on Nuclear Physics 2024 at Indian Institute of Technology, Roorkee, India.
- Title :** Comprehensive Assessment of Heat Loads in Ground-Based Cryogenic Applications.
- Authors :** Damini Singh, **Kapil Saraswat**, Avijit Kumar Ganguly, Venktesh Singh.
- Journal :** [DAE Symp. Nucl. Phys. 68 \(2024\) 1149.](#)
37. **Conference :** *68th* DAE Symposium on Nuclear Physics 2024 at Indian Institute of Technology, Roorkee, India.
- Title :** Boosting sensitivity of ton-scale rare event search experiments via improved thermal shielding of cryostats.
- Authors :** Manoj Kumar Singh, **Kapil Saraswat**, Venktesh Singh, Damini Singh.
- Journal :** [DAE Symp. Nucl. Phys. 68 \(2024\) 1253.](#)

Chapters in Books

1. **Title :** Charged and Neutral coherent pion production in neutrino-nucleus scattering.
 Authors : **Kapil Saraswat**, Prashant Shukla, Vineet Kumar, and Venktesh Singh.
 Series : Springer Proceedings in Physics book series.
 Book : [Springer Proc. Phys. 203](#) (2018) 697, [\[arXiv:1805.11836 \[hep-ph\]\]](#).
2. **Title :** Freeze out of strange hadrons in pp, pPb and PbPb collisions at LHC energies.
 Authors : **Kapil Saraswat**, Prashant Shukla, Vineet Kumar, and Venktesh Singh.
 Series : Springer Proceedings in Physics book series.
 Book : [Springer Proc. Phys. 203](#) (2018) 701, [\[arXiv:1805.11832 \[hep-ph\]\]](#).
3. **Title :** Identified Charged Particle Production in Pb+Pb Collisions at $\sqrt{s_{NN}} = 2.76$ TeV Using Tsallis Distribution Function.
 Authors : P. Kumar, P. K. Khandai, **K. Saraswat**, and V. Singh.
 Series : Springer Proceedings in Physics Book Series : Proceedings of the National Workshop on Recent Advances in Condensed Matter and High Energy Physics.
 Book : [Springer Proc. Phys. 278](#) (2022) 151-156.

List of Attended Schools and Workshops

- **Workshop on High Energy Physics @ Sangam 2013** at Harish Chandra Research Institute, Allahabad.
- **Winter School on High Energy Physics 2013** at Banaras Hindu University, Varanasi.
- **Winter School on Nuclear, Particle and Accelerator Physics 2014** at Banaras Hindu University, Varanasi.
- **Workshop on Light from Dark Side of the Universe 2015** at Banaras Hindu University, Varanasi.
- **Winter School on Beyond the Standard Model Physics 2016** at Banaras Hindu University, Varanasi.
- **A Workshop on Neutrino Physics : Theory and Experiments 2019** at Banaras Hindu University, Varanasi.
- **Taiwan High Energy Physics meeting and Taiwan Instrumentation Detector Consortium 2021** at Tainan, Taiwan.
- **PIRE-GEMADARC Winter Collaboration Meeting-2022 (ONLINE)** at University of South Dakota, USA.

Talks and Poster Presentations

1. **Conference :** ^{57th} DAE BRNS Symposium on Nuclear Physics 2012 at Department of Physics and Astrophysics, University of Delhi, Delhi, India.
Title : Trigger Detector Study for Resistive Plate Chamber Detectors.
Mode : Poster Presentation.
2. **Conference :** ^{58th} DAE BRNS International Symposium on Nuclear Physics 2013 at Bhabha Atomic Research Centre, Mumbai, India.
Title : Study of VECC Prototype INO ICAL using cosmic ray muon and g4beamline.
Mode : Poster Presentation.
3. **Conference :** ^{59th} DAE BRNS Symposium on Nuclear Physics 2014 at Department Of Physics, Banaras Hindu University, Varanasi, India.
Title : Constraining Heavy Quark Energy Loss Using B and D Mesons in Heavy Ion Collision.
Mode : Poster Presentation.
4. **Conference :** ^{59th} DAE BRNS Symposium on Nuclear Physics 2014 at Department Of Physics, Banaras Hindu University, Varanasi, India.
Title : Interaction of neutrinos with matter in low and intermediate energy region.

Mode : Oral Presentation.

5. **Conference :** 60th DAE BRNS Symposium on Nuclear Physics 2015 at Sri Sathya Sai Institute of Higher Learning, Prasanthi Nilayam, India.

Title : Heavy quark energy loss in PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV.

Mode : Poster Presentation.

6. **Conference :** 7th International Conference on Physics and Astrophysics of Quark Gluon Plasma (ICPAQGP) 2015 at Variable Energy Cyclotron Center, Kolkata, India.

Title : Energy loss of B and D mesons in PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV.

Mode : Oral Presentation.

7. **Conference :** 61th DAE BRNS Symposium on Nuclear Physics 2016 at Saha Institute of Nuclear Physics, Kolkata, India.

Title : Strange hadron production in high multiplicity pp collision at $\sqrt{s} = 7$ TeV.

Mode : Poster Presentation.

8. **Conference :** 61th DAE BRNS Symposium on Nuclear Physics 2016 at Saha Institute of Nuclear Physics, Kolkata, India.

Title : Quasi elastic scattering of neutrino with nuclei.

Mode : Oral Presentation.

9. **Conference :** XXII DAE BRNS HIGH ENERGY PHYSICS SYMPOSIUM 2016 at University of Delhi, Delhi, India.

Title : Charged and Neutral coherent pion production in neutrino-nucleus scattering.

Mode : Poster Presentation.

10. **Conference :** XXII DAE BRNS HIGH ENERGY PHYSICS SYMPOSIUM 2016 at University of Delhi, Delhi, India.

Title : Freeze out of strange hadrons in pp, pPb and PbPb collisions at LHC energies.

Mode : Poster Presentation.

11. **Conference :** 62th DAE BRNS Symposium on Nuclear Physics 2017 at Thapar University, Patiala, India.

Title : Heavy quark energy loss and heavy meson spectra in heavy ion collisions at LHC energies.

Mode : Poster Presentation.

12. **Conference :** 62th DAE BRNS Symposium on Nuclear Physics 2017 at Thapar University, Patiala, India.

Title : Charged particle production in pp and heavy ion collisions.

Mode : Poster Presentation.

13. **Conference :** ^{62th} DAE BRNS Symposium on Nuclear Physics 2017 at Thapar University, Patiala, India.
Title : Study of strongly interacting matter using light and heavy hadron production in heavy ion collisions.
Mode : Poster Presentation.
14. **Conference :** ^{63th} DAE International Symposium on Nuclear Physics 2018 at Bhabha Atomic Research Centre, Mumbai, India.
Title : Identified charged particle production in proton-proton collisions at $\sqrt{s} = 13$ TeV.
Mode : Poster Presentation.
15. **Conference :** ^{63th} DAE International Symposium on Nuclear Physics 2018 at Bhabha Atomic Research Centre, Mumbai, India.
Title : Nuclear modification factor and energy loss of charged particles and jets in PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV.
Mode : Poster Presentation.
16. **Conference :** Symposium on Advances in Physics from Large to Small Scales (APSLS) 2019 at Department of Physics (UGC-CAS) DSB Campus, Kumaun University, Nainital, India.
Title : Heavy Quark Energy Loss in PbPb Collisions at $\sqrt{s_{NN}} = 5.02$ TeV.
Mode : Oral Presentation.
17. **Conference :** ^{64th} DAE BRNS Symposium on Nuclear Physics 2019 at University of Lucknow, Lucknow, India.
Title : Heavy quark energy loss and heavy meson spectra in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV.
Mode : Oral Presentation.
18. **Conference :** ^{64th} DAE BRNS Symposium on Nuclear Physics 2019 at University of Lucknow, Lucknow, India.
Title : Centrality dependence of Identified charged Particle Production in Pb+Pb collisions at $\sqrt{s} = 2.76$ TeV.
Mode : Poster Presentation.
19. **Conference :** ^{64th} DAE BRNS Symposium on Nuclear Physics 2019 at University of Lucknow, Lucknow, India.
Title : Deuteron production in Au + Au collisions at RHIC energies.
Mode : Poster Presentation.
20. **Conference :** ^{65th} DAE BRNS Symposium on Nuclear Physics 2021 at Bhabha Atomic Research Centre, Mumbai, India.

Title : Effect of energy loss on nuclear modification factor of heavy mesons in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV.

Mode : Oral Presentation.

References

1. Prof. Venktesh Singh
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Webpage

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<https://vidwan.inflibnet.ac.in/profile/448426>.
2. **ORCID iD** 
<https://orcid.org/my-orcid?orcid=0000-0002-1211-5652>.
3. **SCOPUS** 
<https://www.scopus.com/authid/detail.uri?authorId=56825317400>.
4. **arXiv** 
<https://arxiv.org/search/hep-ph?searchtype=author&query=Saraswat%2C+K>.
5. **iNSPIRE-HEP** 
<https://inspirehep.net/authors/1191726>.
6. **Google Scholar** 
<https://scholar.google.com/citations?user=paum46oAAAAJ&hl=en&oi=sra>.
7. **ResearchGate**  
<https://www.researchgate.net/profile/Kapil-Saraswat>.